DataPilot AI - Data Science Report

Automated Machine Learning Analysis

1. Dataset Overview

File Name	employee_attrition.csv
Rows	20
Columns	11
Numeric Columns	4
Categorical Columns	7

2. Dataset Understanding

Summary: The dataset contains information about employees, including their ID, age, salary, years at the company, gender, department, education level, work-life balance, overtime status, marital status, and attrition status. It has 20 rows and 11 columns.

Problem Type: Classification

Observations:

• The dataset is small with only 20 rows, which might limit the model's performance.

• All numeric columns are integers, while categorical columns are strings, ensuring consistent data type handling in preprocessing steps.

• There are no missing or duplicate values, suggesting a clean and well-maintained dataset.

3. Data Quality

Metric	Value
Duplicate Rows	0
Duplicate Percentage	0.00%
Constant Columns	0
Empty Columns	0

4. Data Cleaning

Metric	Value
Original Rows	20
Final Rows	20

Duplicate Rows Removed	0
Columns Removed	0
Missing Value Columns Handled	0

Cleaning Summary:

• Trimmed whitespace from string columns.

5. Analysis Plan

Target Column	Attrition
Problem Type	Classification
Evaluation Metric	f1_score
Train/Test Split	0.2
Random State	42
Scaling	standard
Feature Encoding	one_hot
Target Encoding	label

Numerical Features: Age, Salary, YearsAtCompany

Categorical Features: Gender, Department, Education, WorkLifeBalance, OverTime, MaritalStatus

6. Exploratory Data Analysis

Numerical Summary

Age

Statistic	Value
count	20.000
mean	33.500
std	6.460
min	24.000
25%	28.750
50%	32.500
75%	38.250
max	45.000

Salary

Statistic	Value

count	20.000
mean	68500.000
std	18343.219
min	43000.000
25%	55750.000
50%	63000.000
75%	82500.000
max	102000.000

YearsAtCompany

Statistic	Value
count	20.000
mean	6.750
std	4.459
min	1.000
25%	3.000
50%	5.500
75%	10.000
max	16.000

Categorical Summary

Gender

Category	Count
Male	10
Female	10

Department

Category	Count
IT	7
Sales	5
HR	4
Finance	4

Education

Category	Count
Bachelor	9
Master	8

PhD	3

WorkLifeBalance

Category	Count
Good	9
Excellent	7
Fair	4

OverTime

Category	Count
No	12
Yes	8

MaritalStatus

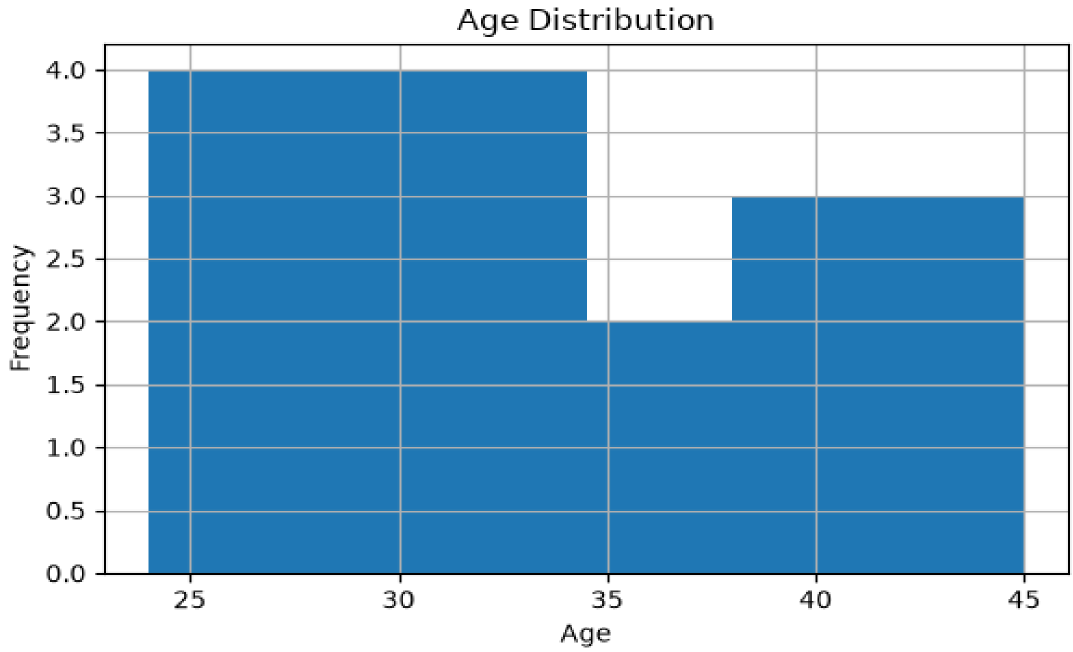
Category	Count
Married	10
Single	8
Divorced	2

Target Distribution

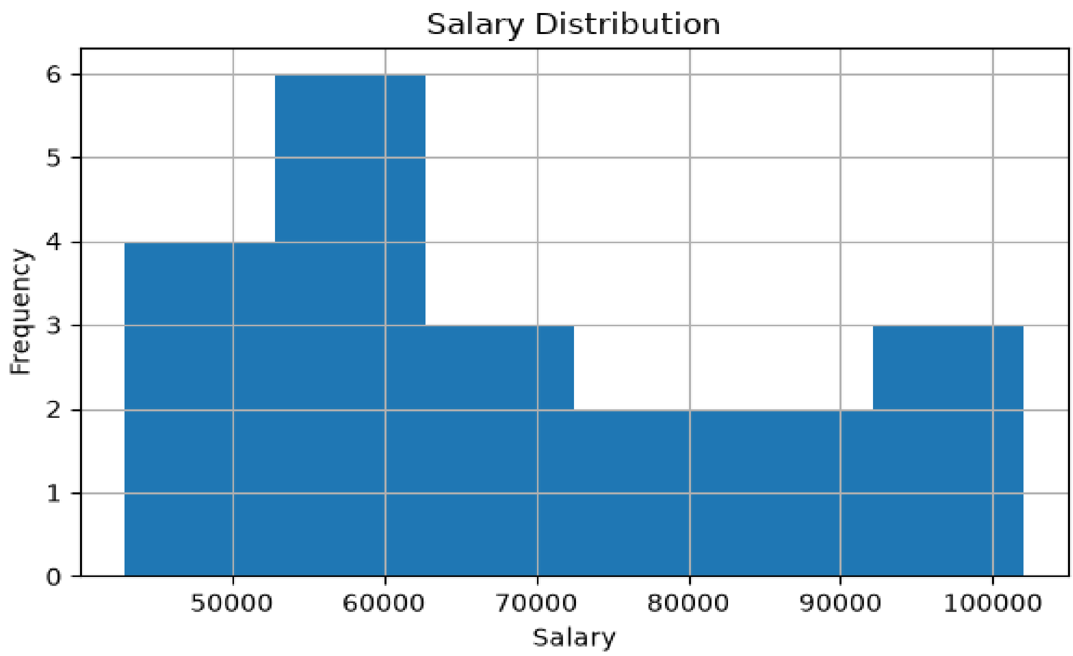
Target	Count
No	12
Yes	8

7. EDA Visualizations

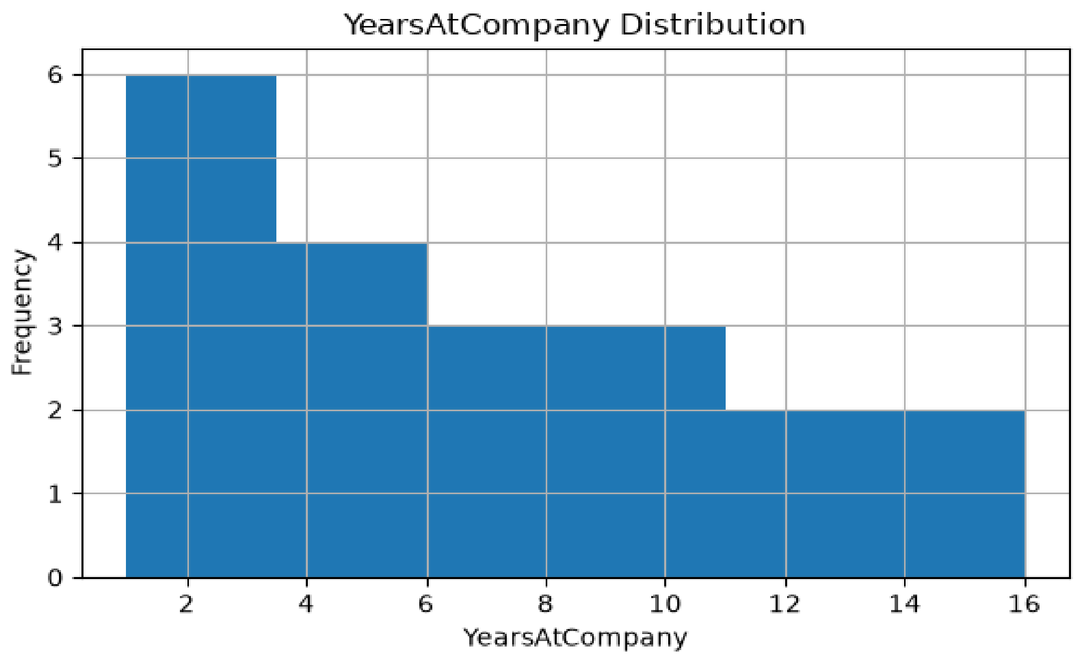
Age Histogram



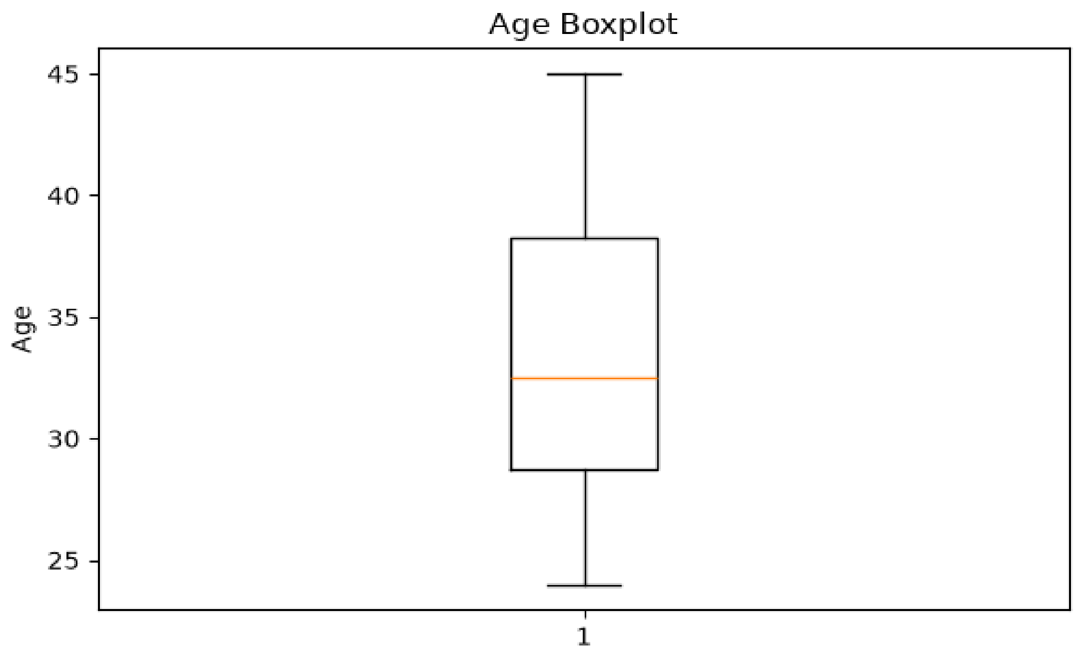
Salary Histogram



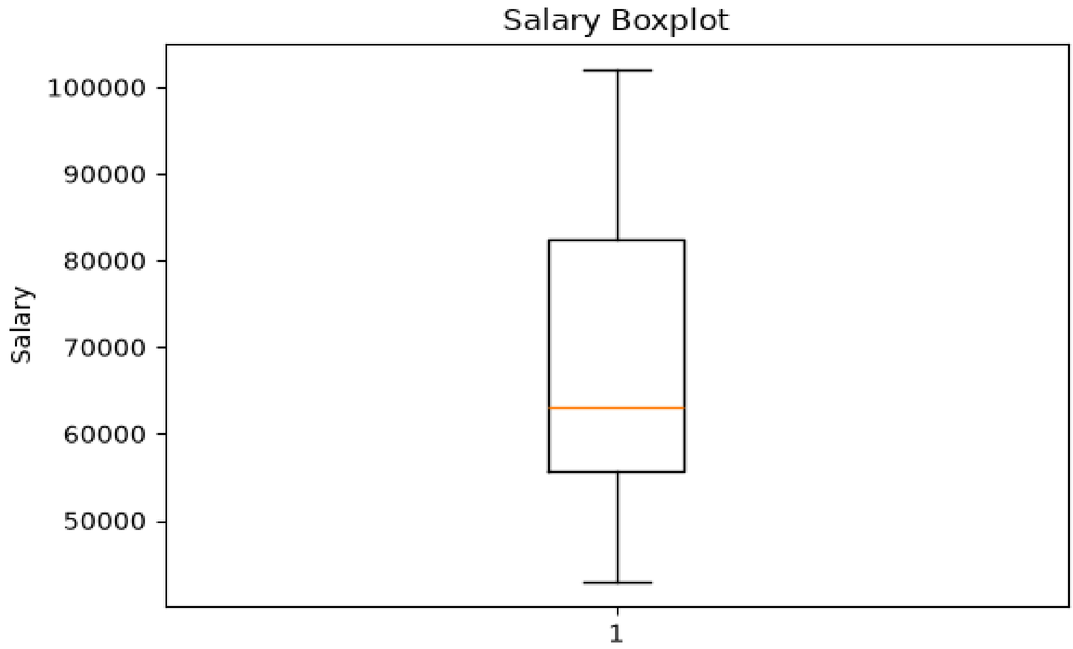
Yearsatcompany Histogram



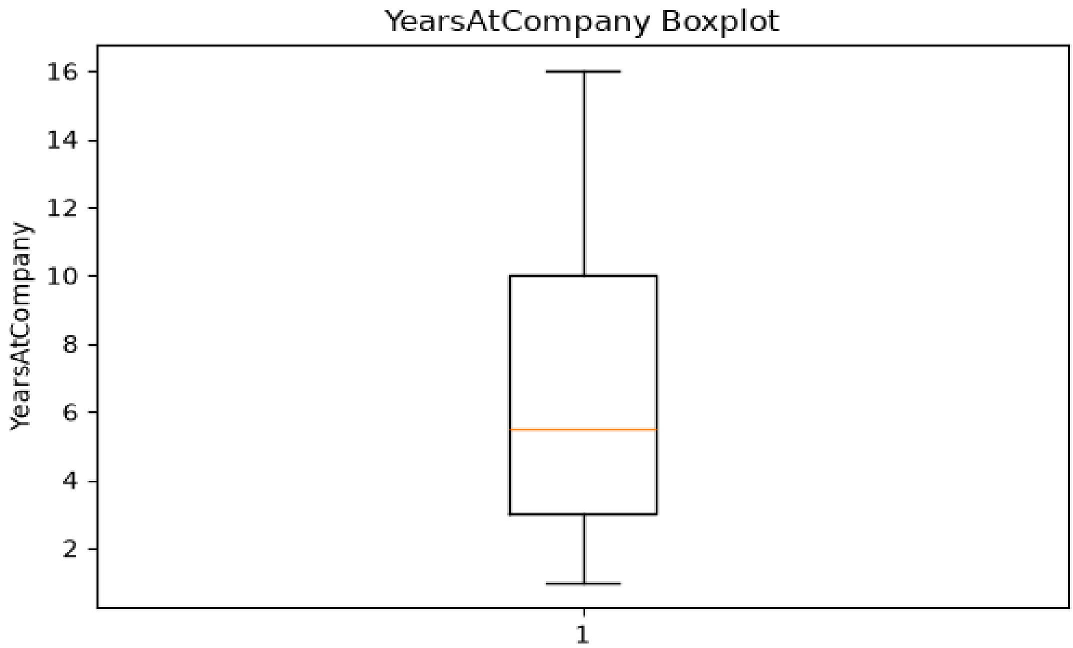
Age Boxplot



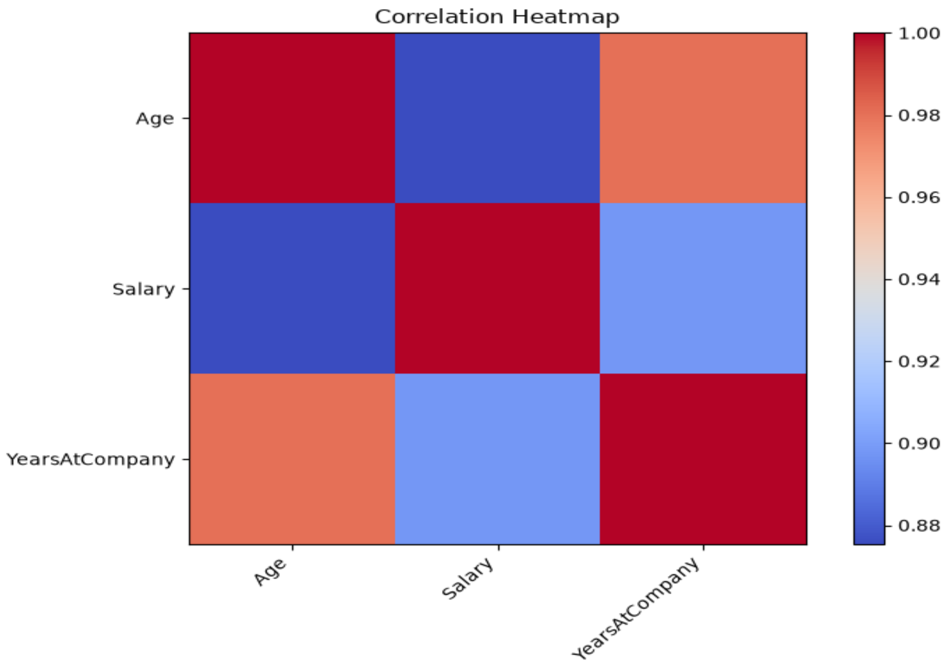
Salary Boxplot



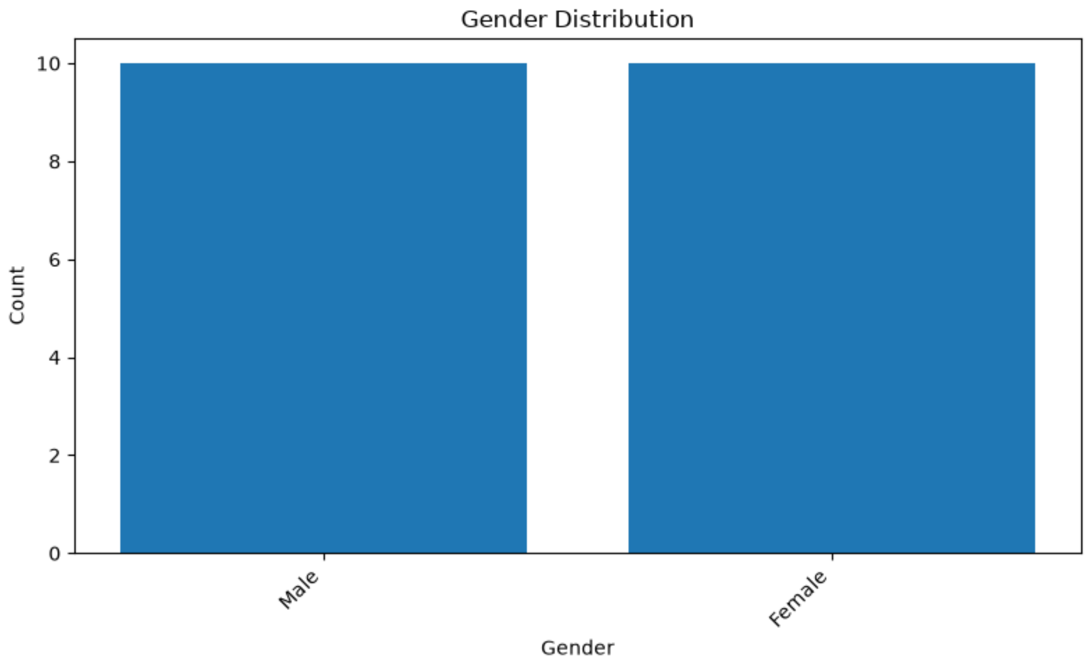
Yearsatcompany Boxplot



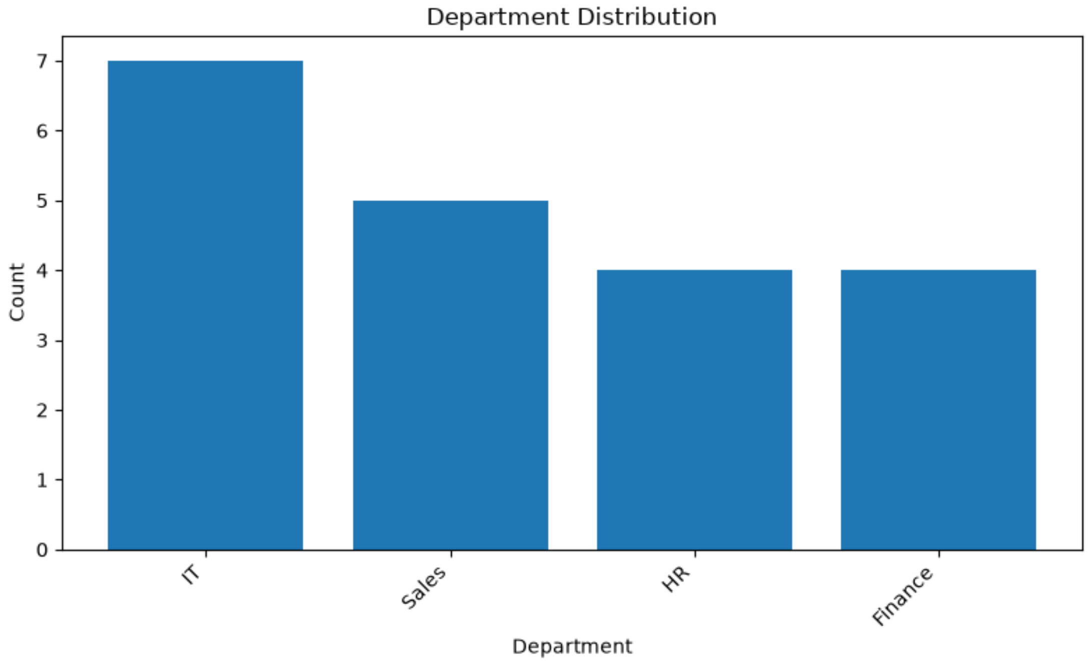
Correlation Heatmap



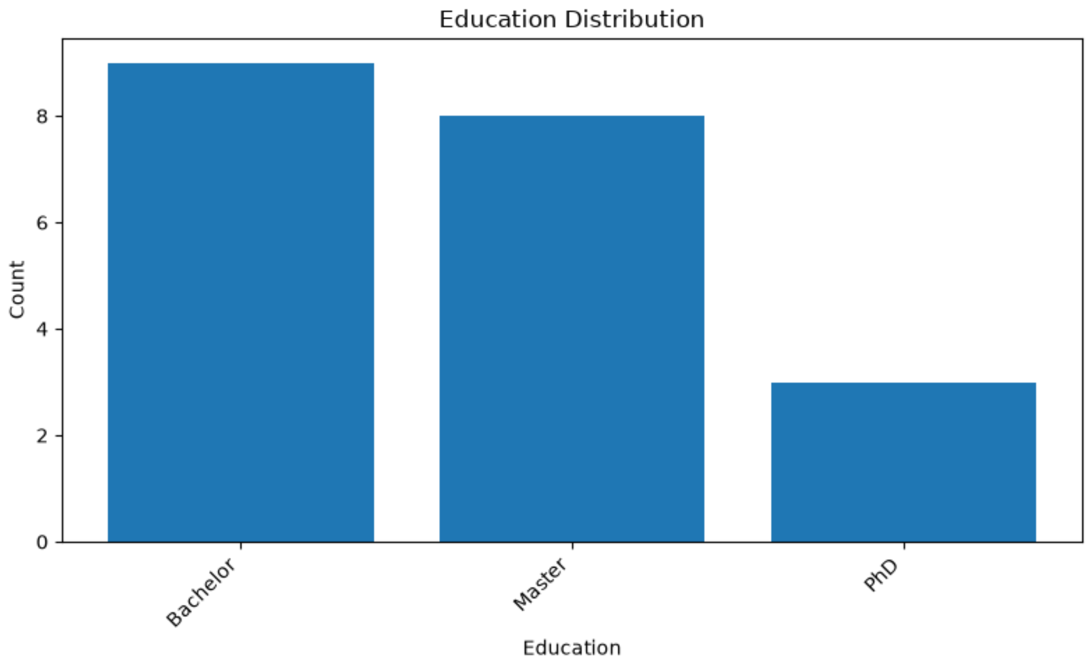
Gender Countplot



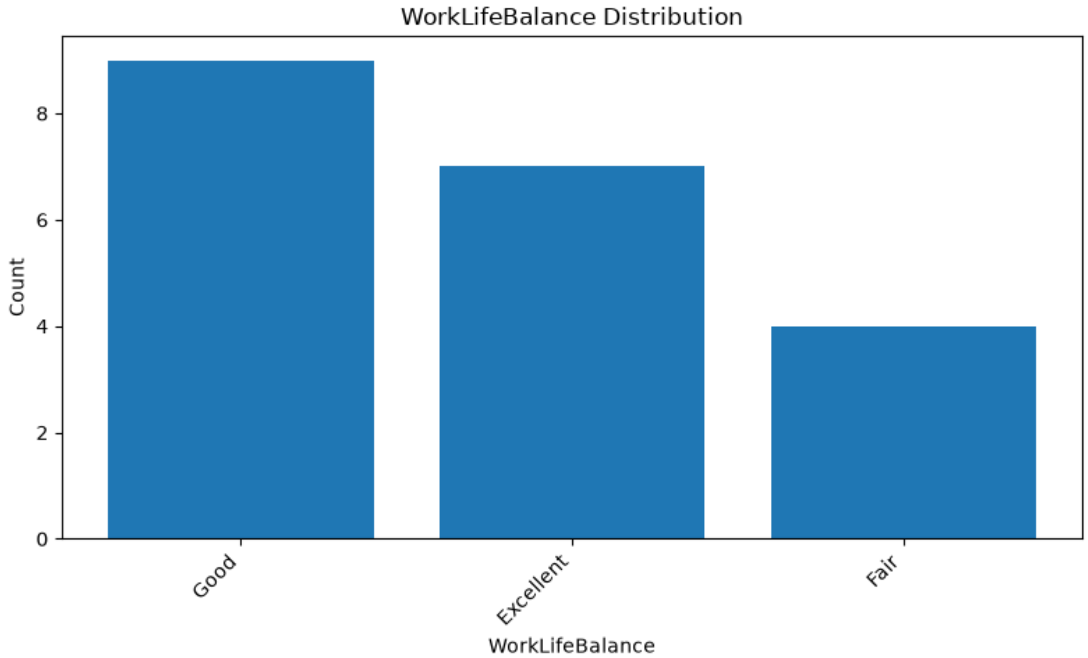
Department Countplot



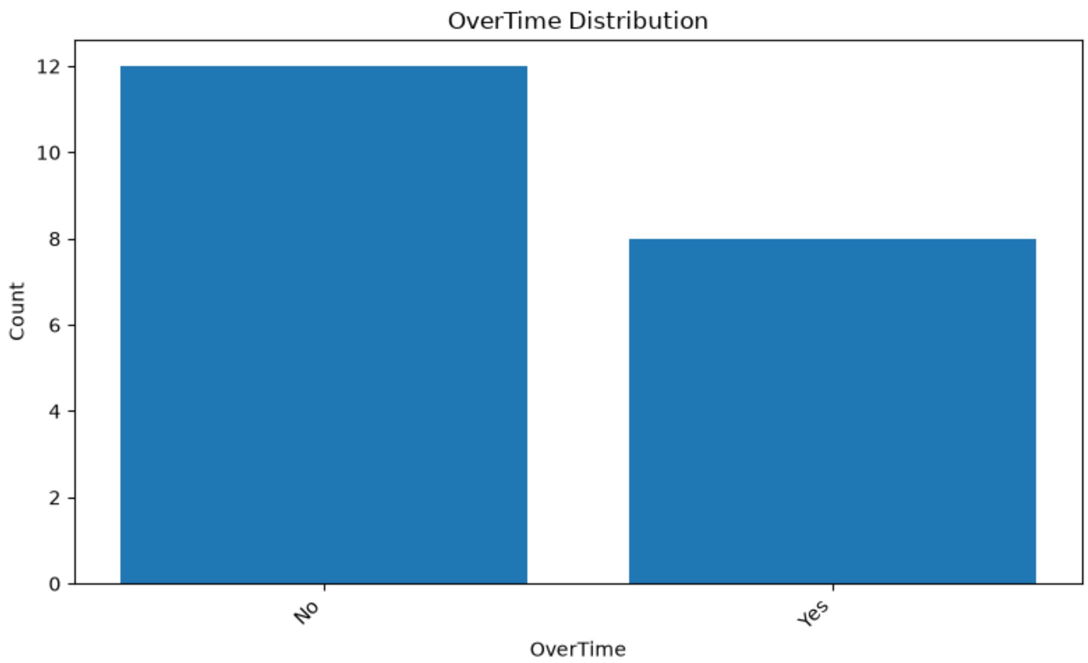
Education Countplot



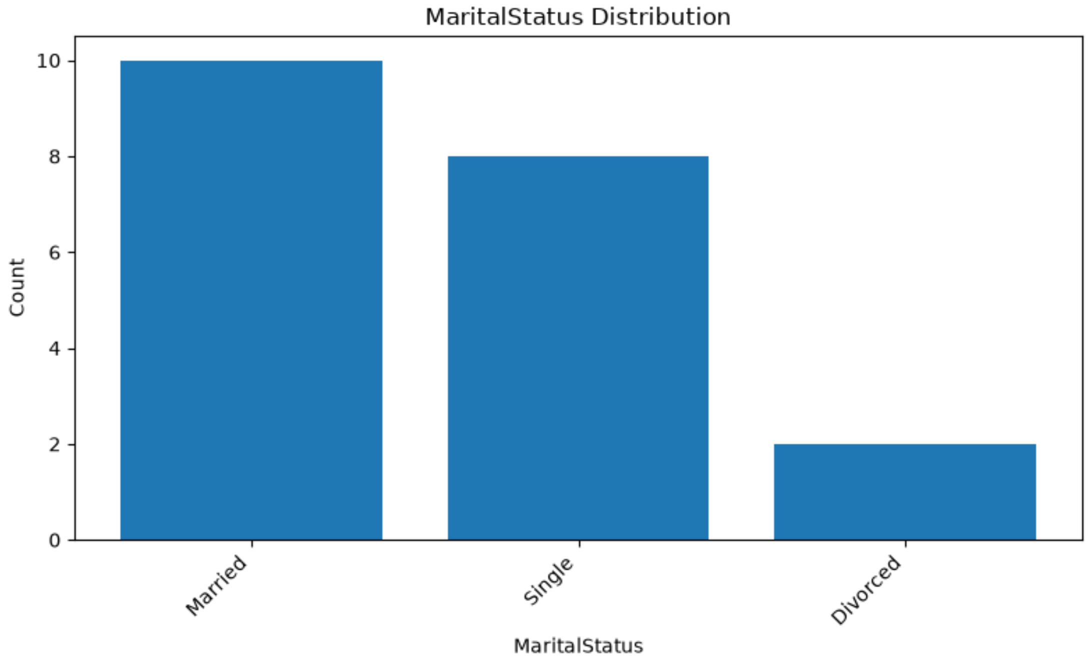
Worklifebalance Countplot



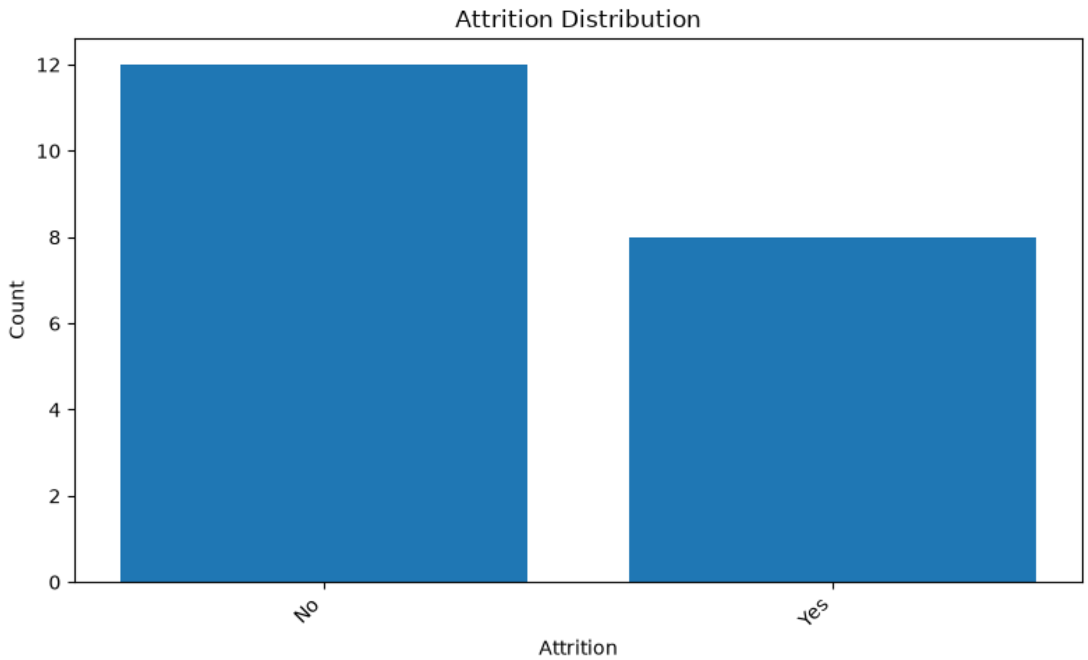
Overtime Countplot



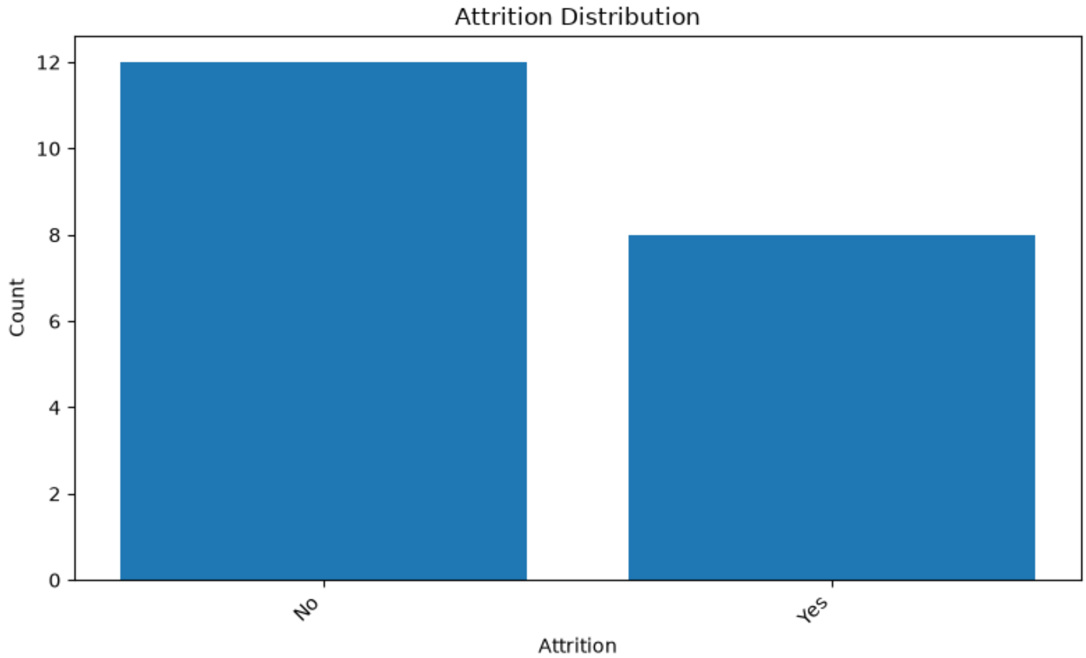
Maritalstatus Countplot



Attrition Countplot



Target Distribution



8. Feature Engineering

Total Features: 20

Feature Matrix Shape: (20, 20)

Generated Features:

• Age

• Salary

• YearsAtCompany

• Gender_Female

• Gender_Male

• Department_Finance

• Department_HR

• Department_IT

• Department_Sales

• Education_Bachelor

• Education_Master

• Education_PhD

• WorkLifeBalance_Excellent

• WorkLifeBalance_Fair

• WorkLifeBalance_Good

• OverTime_No

• OverTime_Yes

• MaritalStatus_Divorced

• MaritalStatus_Married

• MaritalStatus_Single

9. Model Training

Model	Score
logistic_regression	1.0000
random_forest	1.0000

Best Model: logistic_regression

Best Score: 1.0000

Training Samples: 16

Testing Samples: 4

10. Model Evaluation

Metric	Value
accuracy	1.0000
precision	1.0000
recall	1.0000
f1_score	1.0000

Confusion Matrix

Actual / Predicted	0	1
0	2	0
1	0	2

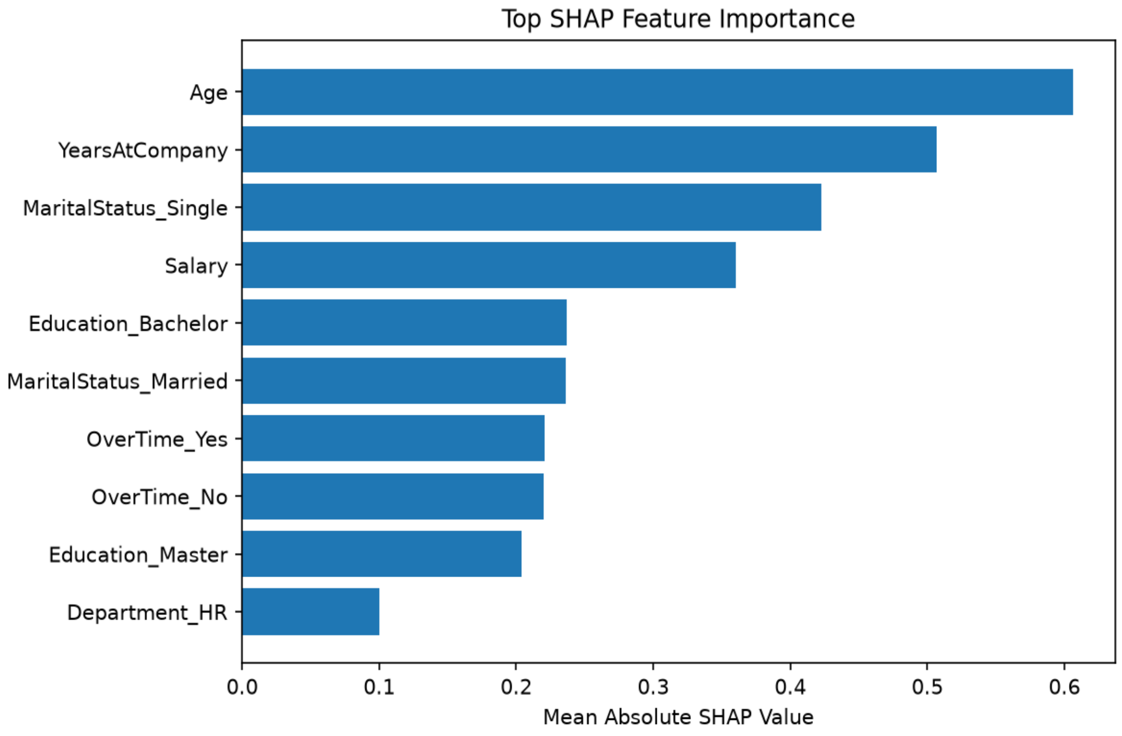
11. Model Explainability

Top Features

Feature	Importance
Age	0.6065
YearsAtCompany	0.5068
MaritalStatus_Single	0.4223
Salary	0.3600
Education_Bachelor	0.2368
MaritalStatus_Married	0.2364
OverTime_Yes	0.2204
OverTime_No	0.2201
Education_Master	0.2037

Department_HR	0.0999

SHAP Feature Importance



12. Business Insights

Executive Summary

The dataset contains information about employees and has 20 rows, which is a small sample size. All data types are consistent, with no missing or duplicate values, indicating clean and well-maintained data. The exploratory analysis reveals that the correlation between numerical features (like Age, YearsAtCompany, Salary) is strong, and there are significant categorical features such as MaritalStatus_Single and OverTime_Yes/No impacting the model. Despite the small dataset size, a machine learning model achieved perfect accuracy, precision, recall, and F1-score in the classification task. The SHAP values provide insights into feature importance, with Age and YearsAtCompany being critical factors for predicting attrition status.

Key Findings

• The model performance is excellent, achieving 100% accuracy, precision, recall, and F1-score.

• Key features influencing the prediction include Age, YearsAtCompany, MaritalStatus_Single, OverTime_Yes/No, Salary, and others.

• There is a high correlation between numerical features (Age, YearsAtCompany, Salary) which might indicate redundancy or need for feature engineering.

Model Performance Summary

The model demonstrates perfect performance on the given dataset with all metrics at 100%. This suggests that either the data is highly indicative of the target variable or the model has overfit the training data. Given the small size of the dataset, this may not generalize well to a larger population.

Feature Importance Summary

Age and YearsAtCompany are significant features with strong positive impacts on predicting attrition status. MaritalStatus_Single also plays a crucial role in determining whether an employee is likely to leave. Features like OverTime_Yes/No, Salary, and others show varying levels of importance as well, but their impact needs further validation outside this specific dataset.

Business Recommendations

• Due to the high accuracy, consider implementing the model immediately for decision-making purposes, but monitor its performance closely over time.

• Conduct further feature analysis and possibly add more features or gather additional data points if available, especially considering the limited sample size.

• Implement a cross-validation strategy to better understand how well the model generalizes outside this specific dataset.

• Use the insights from SHAP values to prioritize areas for potential improvement in employee retention strategies.

Risks and Limitations

• The small dataset size may lead to overfitting and poor generalizability. Further data collection or augmentation is recommended.

• There are strong correlations between features, which might indicate redundancy or the need for dimensionality reduction techniques like PCA.

• Given the high accuracy on this specific dataset, it’s crucial to validate the model's performance using a different testing set or through cross-validation.

Next Steps

• Collect more data points and include additional features relevant to employee attrition prediction.

• Implement feature selection and engineering techniques to improve model robustness and avoid overfitting.

• Deploy the model in a pilot environment for real-world testing and gather feedback from stakeholders.

• Regularly update and retrain the model with new data to ensure its relevance and accuracy.